

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + bx_2^2 + bx_3^2) + 2x_2x_3.$$

- (a) (3 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & ab & 1 \\ 0 & 1 & ab \end{pmatrix}$$

(1 marks) Its third order principal minor is $a^3b^2 - a$. Its second order principal minors are a^2b and $a^2b^2 - 1$. Its first order principal minors are a and ab .

- (b) (2 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*.

Solutions: A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$.

- (c) (15 marks) Find the values of the parameter values, a and b , such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? Then, plot these regions in $\mathbb{R}^2$.

Solutions: (4 Marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$a \geq 0, \quad ab \geq 0, \quad a^2b \geq 0, \quad a^2b^2 - 1 \geq 0, \quad a^3b^2 - a \geq 0.$$

For these to hold at once it must be that $b > 0$ and $a \geq 1/b$.

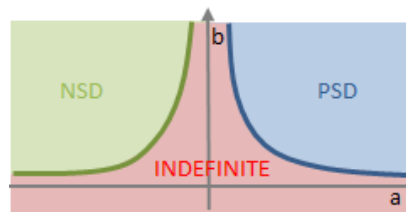
(4 Marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

$$a \leq 0, \quad ab \leq 0, \quad a^2b \geq 0, \quad a^2b^2 - 1 \geq 0, \quad a^3b^2 - a \leq 0.$$

For these to hold at once it must be that $b > 0$ and $a \leq -1/b$.

(2 Marks) $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite. Thus, $Q(\mathbf{x})$ is indefinite whenever either $b \leq 0$ or $b > 0$ and $a \in (-1/b, 1/b)$.

(5 Marks) The plot below depicts the answer



2. (20 marks) Consider a function $f : U \rightarrow \mathbb{R}$ for some convex set U in $\mathbb{R}^n$, and answer the following questions:

- (a) (14 marks) Prove that if $f(\mathbf{x})$ is *quasi-concave*, then the function $h(\mathbf{x}) = g(f(\mathbf{x}))$ is *quasi-convex* for any *strictly decreasing* transformation $g : \mathbb{R} \rightarrow \mathbb{R}$.

Solutions: Let f be quasi-concave and defined on a convex set U in $\mathbb{R}^n$. By assumption the transformation $g : \mathbb{R} \rightarrow \mathbb{R}$ is strictly decreasing, meaning that $g' < 0$. For any function $u : \mathbb{R}^n \rightarrow \mathbb{R}$, let $C_k^+(u)$ denote its upper-contour sets and let $C_k^-(u)$ denote its lower-contour sets. So if $h(\mathbf{x}) = g(f(\mathbf{x}))$, we have that

$$C_k^+(f) = \{\mathbf{x} \in U | f(\mathbf{x}) \geq k\} = \{\mathbf{x} \in U | h(\mathbf{x}) \leq g(k)\} = C_{g(k)}^-(h).$$

The latter follows because by $g' < 0$ we have that

$$f(\mathbf{x}) \geq k \Leftrightarrow g(f(\mathbf{x})) \leq g(k).$$

But since $C_k^+(f)$ is convex for all k given that f is quasi-concave, $C_{g(k)}^-(h)$ is convex for all $g(k)$. Hence h is quasi-convex.

- (b) (6 marks) Let $U = \mathbb{R}_{++}^3$, consider the function

$$h(\mathbf{x}) = -\exp(x_1^2 x_2^4 x_3^6),$$

Is the function *quasi-convex*?

Solutions: We can exploit the previous observation to argue immediately that $h(\mathbf{x})$ is quasi-convex. Consider the strictly decreasing transformation $g^{-1}(k) = \log(\log(-k))$. If so,

$$f(\mathbf{x}) = g^{-1}(h(\mathbf{x})) = 2 \log x_1 + 4 \log x_2 + 6 \log x_3.$$

Moreover f is obviously quasi-concave since it is a sum of concave functions. Therefore, $h(\mathbf{x})$ is quasi-convex since $h(\mathbf{x}) = g(f(\mathbf{x}))$ for the strictly decreasing transformation $g(k) = -\exp(\exp(k))$.

3. (40 marks) For $a > 0$, consider the problem:

$$\max_{x,y} y^{1/2} x^{1/2} \quad \text{s.t.} \quad y + ax \leq 1/4, \quad y \geq 0, \quad x \geq a.$$

- (a) (6 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the Kuhn-Tucker method?

Solutions: (2 marks) The objective function is concave, since its Hessian is negative semidefinite on the constraint set since for $y > 0$ and $x \geq a$

$$D^2 f(x, y) = \begin{pmatrix} -y^{1/2} x^{-3/2} / 4 & y^{-1/2} x^{-1/2} / 4 \\ y^{-1/2} x^{-1/2} / 4 & -y^{-3/2} x^{1/2} / 4 \end{pmatrix},$$

$$|D^2 f(x, y)| = 0 \geq 0, \quad -y^{1/2} x^{-3/2} / 4 \leq 0, \quad -y^{-3/2} x^{1/2} / 4 \leq 0.$$

The Hessian is not defined for $y = 0$. (1 mark) The constraints are linear and thus convex. (3 marks) Thus, when the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a \in (0, 1/2)$, since the point $\tilde{x} = a + \varepsilon$ and $\tilde{y} = \varepsilon$ is always in the interior of the constraint set for $\varepsilon \in (0, (1/4 - a^2)/(1 + a))$, since

$$\tilde{y} + a\tilde{x} = a^2 + (1 + a)\varepsilon < 1/4, \quad \tilde{x} > a, \quad \tilde{y} > 0,$$

which in turn requires $1/4 > a^2$ or equivalently $a < 1/2$.

- (b) (6 marks) Suppose the restrictions on a you found in first part of the problem hold. For $y > 0$, write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?

Solutions: (5 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= (x/y)^{1/2}/2 - \lambda_1 + \lambda_2, \\ 0 &= (y/x)^{1/2}/2 - a\lambda_1 + \lambda_3, \\ 0 &= \lambda_1(y + ax - 1/4) = \lambda_2 y = \lambda_3(x - a), \\ 0 &\geq y + ax - 1/4, \quad 0 \geq -y, \quad 0 \geq a - x, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(1 marks) Any solutions to these conditions is maximizer of the Lagrangian, because the Lagrangian is a sum of concave functions by the arguments in (a) and because Slater condition holds.

- (c) (20 marks) Suppose the restrictions on a you found in first part of the problem hold. Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (2 marks) At a maximum it must be that $\lambda_2(a) = 0$. If we had that $\lambda_2(a) > 0$ and $y(a) = 0$, the value of the objective could be increased while still fulfilling all the constraints. If $x(a) < 1/4a$, this could be accomplished by setting $\tilde{y} = e$ such that $e < 1/4 - ax$ and keeping fixed x when since

$$f(x(a), 0) = 0 < f(x(a), \tilde{y}).$$

When $x(a) = 1/4a > a$, this could be accomplished by setting $\tilde{y} = e$ and $\tilde{x} = 1/4a - e$, for $e < 1/4a - a$ so to guarantee $\tilde{x} > a$, since

$$f(x(a), 0) = 0 < f(\tilde{x}, \tilde{y}).$$

- (2 marks) At a maximum, it must be that $\lambda_1(a) > 0$. If $\lambda_1(a) = \lambda_2(a) = 0$, the first order KKT condition for variable y would require $x(a) = 0$ which would be inconsistent with the constraint requiring $x(a) \geq a > 0$.
- (8 marks) Next, consider the case in which $\lambda_1(a) > 0$ while $\lambda_2(a) = \lambda_3(a) = 0$. If so, solving the two first order conditions and the binding constraint for variables x , y and λ_1 would imply that

$$x(a) = \frac{1}{8a} \geq a, \quad y(a) = \frac{1}{8} \geq 0, \quad \lambda_1(a) = \left(\frac{1}{4a}\right)^{1/2} \geq 0,$$

which requires $a \leq 1/\sqrt{8}$ to ensure that $x(a) \geq a$.

- (8 marks) Finally, consider the case in which $\lambda_1(a), \lambda_3(a) > 0$, while $\lambda_2(a) = 0$. If so, the two binding constraints would imply that

$$y(a) = 1/4 - a^2 \quad \text{and} \quad x(a) = a.$$

Solving the two first order conditions for variables x and y would imply that

$$\lambda_1(a) = \frac{1}{2} \left(\frac{a}{1/4 - a^2} \right)^{1/2}, \quad \lambda_3(a) = \frac{a}{2} \left(\frac{a}{1/4 - a^2} \right)^{1/2} - \frac{1}{2} \left(\frac{1/4 - a^2}{a} \right)^{1/2}.$$

But then $\lambda_3(a) \geq 0$ would require $a \geq 1/\sqrt{8}$.

- (d) (8 marks) Suppose the restrictions on a you found in first part of the problem hold. Use the Envelope Theorem to find the slope of value function with respect to the parameter a , and discuss its sign.

Solutions: (1 marks) The Envelope Theorem states that if $(x(a), y(a), \lambda(a))$ are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \frac{\partial L(x(a), y(a), \lambda(a))}{\partial a} = -\lambda_1(a)x(a) - \lambda_3(a).$$

(3 marks) If $a \in (0, 1/\sqrt{8})$, then

$$\frac{dv(a)}{da} = -\left(\frac{1}{a}\right)^{1/2} \frac{1}{16a} < 0.$$

Intuitively, the slope of the value function is decreasing as the constraint set shrinks.

(4 marks) If $a \in (1/\sqrt{8}, 1/2)$, then

$$\begin{aligned} \frac{dv(a)}{da} &= \frac{1}{2} \left(\frac{1/4 - a^2}{a} \right)^{1/2} - a \left(\frac{a}{1/4 - a^2} \right)^{1/2} \\ &= \left(\frac{1 - 4a^2}{4a} \right)^{1/2} \left(\frac{1}{2} - \frac{4a^2}{1 - 4a^2} \right) < 0, \end{aligned}$$

where the inequality holds whenever $\frac{1}{2} - \frac{4a^2}{1-4a^2} < 0$ which is equivalent to requiring $a > 1/\sqrt{12}$ which must hold since $a > 1/\sqrt{8}$. Intuitively, the slope of the value function is decreasing as the constraint set shrinks for two reasons now.

4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 1]^2 \rightrightarrows [0, 1]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 \in (a, 1] \\ [0, 1] & \text{if } x_2 = a \\ 0 & \text{if } x_2 \in [0, a) \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [1/2, 1] & \text{if } x_1 = 1 \\ 1/2 & \text{if } x_1 \in (a, 1) \\ [0, 1/2] & \text{if } x_1 = a \\ 0 & \text{if } x_1 \in [0, a) \end{cases},$$

for a scalar $a \in (0, 1)$. Briefly, argue why Kakutani fixed point theorem applies. Then, for all values of $a \in (0, 1)$ find all the fixed points of this correspondence.

Solutions: (2 marks) Both correspondences have closed graph and are therefore UHC. Moreover, both are obviously convex-valued and non-empty. Since the image of $\mathbf{h}$ belongs to $[0, 1]^2$ which is also the domain, the Fixed Point Theorem applies. Thus, there will be at least one fixed point on $[0, 1]^2$.

(6 marks) First suppose that $a < 1/2$. If so, the set of fixed points of $\mathbf{h}$ consists of: $(0, 0)$, (a, a) , and $(1, x_2)$ for any $x_2 \geq 1/2$. The point $(0, 0)$ is a fixed point since

$$(0, 0) = \mathbf{h}(0, 0).$$

The point (a, a) is a fixed point since

$$(a, a) \in [0, 1] \times [0, 1/2] = \mathbf{h}(a, a).$$

The points $(1, x_2)$ for any $x_2 \in [1/2, 1]$ are fixed points since

$$(1, x_2) \in \{1\} \times [1/2, 1] = \mathbf{h}(1, x_2).$$

Moreover, no other point can be a fixed point if $a < 1/2$, because for $x_2 \in (0, a)$ there can be no fixed point since

$$h_1(\mathbf{x}) = 0 \quad \text{and} \quad h_2(0, x_2) = 0,$$

while for $x_2 \in (a, 1/2)$ there can be no fixed point since

$$h_1(\mathbf{x}) = 1 \quad \text{and} \quad h_2(1, x_2) = [1/2, 1].$$

(6 marks) Next suppose that $a = 1/2$. If so, the set of fixed points of $\mathbf{h}$ consists of: $(0, 0)$, $(x_1, 1/2)$ for any $x_1 \geq 1/2$, and $(1, x_2)$ for any $x_2 \geq 1/2$. The point $(0, 0)$ is a fixed point since

$$(0, 0) = \mathbf{h}(0, 0).$$

The points $(x_1, 1/2)$ for any $x_1 \in [1/2, 1)$ are fixed point since

$$(x_1, 1/2) \in [0, 1] \times \{1/2\} = \mathbf{h}(x_1, 1/2).$$

The points $(1, x_2)$ for any $x_2 \in [1/2, 1]$ are fixed points since

$$(1, x_2) \in \{1\} \times [1/2, 1] = \mathbf{h}(1, x_2).$$

Moreover, no other point can be a fixed point if $a = 1/2$, because for $x_2 \in (0, 1/2)$ we have that

$$h_1(\mathbf{x}) = 0 \quad \text{and} \quad h_2(0, x_2) = 0.$$

(6 marks) Finally, suppose that $a > 1/2$. If so, the set of fixed points of $\mathbf{h}$ consists of: $(0, 0)$, and $(1, x_2)$ for any $x_2 \geq a$. The point $(0, 0)$ is a fixed point since

$$(0, 0) = \mathbf{h}(0, 0).$$

The points $(1, x_2)$ for any $x_2 \in [a, 1]$ are fixed points since

$$\begin{aligned} (1, x_2) &\in \{1\} \times [1/2, 1] = \mathbf{h}(1, x_2) \quad \text{for } x_2 > a; \\ (1, a) &\in [0, 1] \times [1/2, 1] = \mathbf{h}(1, a) \end{aligned}$$

Moreover, no other point can be a fixed point if $a > 1/2$, because for $x_2 \in (0, a)$ we have that

$$h_1(\mathbf{x}) = 0 \quad \text{and} \quad h_2(0, x_2) = 0.$$